

1 Training Job

Sparse MoE Distributed Training (3 DP × 4 Pipeline)

S0 S1 S2 S3



 Healthy worker

 Failed worker (rank r)

2 Failure Detection + Safe-Point Repair (R1)

Failure Event
event <r, t, c>



Repair Controller
(Safe-Point Guard)

- mark r RECOVERING
- block opt commit
- discard pre-update work

r : failed rank
t : last committed step
c : checkpoint step

3 R2 Admission Design

Run Risk

$$R_{m,t} = \frac{U[P(Y > 1 | H)]}{\alpha_{run}}$$

Inputs

- run history
- peerAvail
- route / drift
- optimizer

Budgets

- $\eta_f = 0.5\%$
- $\eta_p = 1\%$
- $\alpha_{run} = 5\%$

YES

Guards pass?
 $R \leq 1$?

NO

4a Hybrid Recovery (fast path)

Healthy peers
(step t)



Peer state

Replacement
Rank r



Checkpoint Shard
(step c)



Checkpoint state

Optional Megatron Two-Phase

Phase A:
weights first

forward/backward
updates held

Phase B:
optimizer later

Reintegration (Guarded State Machine)

RECOVERING
(recovering r)

REPAIRED
(weights ready)

BARRIER
(topology ready)

HEALTHY
(fully active)

4b Checkpoint Restart (fallback)

Full Checkpoint
(step c)

Reload
All Ranks

Replay
t - c

Resume
Training

5 Observability Logs (R3)



Auditable
recovery
decisions



policy inputs +
reason



T_{FB} / T_{ready}
 T_{step} latency



state-machine
trace



quality metrics
(joined later)

Control Flow

Path P (Peer)

Path C (Checkpoint)

Hybrid Path

Restart Path